



CHAPTER 3

Reviewing the Literature

Reading publications related to the selected research question prepares the researcher to begin designing a new study that will fill a gap in the literature.

3.1 Informal Sources

A starting point for learning about potential areas of inquiry is to read nontechnical documents and other files available on the Internet. Many major public health organizations, such as the World Health Organization (WHO) and the U.S. Centers for Disease Control and Prevention (CDC), have online factsheets about various diseases and risk factors for disease. Other government agencies and nonprofit organizations also have factsheets, brochures, and websites that provide basic demographic, political, economic, geographic, and other health-related information about countries and regions. Newspapers and popular magazines may present compelling nontechnical articles about exposures, diseases, and populations that highlight what is interesting and important to know about a topic. The websites of patient advocacy groups, personal websites, and other media may also be helpful in identifying and refining an important and meaningful study question.

These initial background readings can provide a foundation for understanding the more technical scientific literature that will be read later as part of a thorough literature review.

However, informal sources that have not been peer reviewed are not part of the formal scientific literature. Researchers must be cautious about any claims in these files that contradict more formal sources of scientific information. Informal sources should never be cited in formal reports and research manuscripts.

3.2 Statistical Reports

When defining specific exposures, diseases, and populations of interest, it may be helpful to identify relevant statistics, such as the estimated prevalence of an exposure in a particular country, the annual global incidence of a disease, or the demographics of a selected population.

For demographic, socioeconomic, and environmental data, reports from national governments are often a good source of up-to-date information. A diversity of statistical estimates are available for every member nation of the United Nations via the World Bank's World Development Indicators reports and the annexes of the annual reports issued by United Nations agencies, such as WHO's *World Health Statistics*.

For health-specific data, national health agencies may publish routine epidemiological updates and reports about **vital statistics**, which are population-level measurements related to births, deaths, and other demographic characteristics. For data about states, provinces, counties, cities, and other smaller governmental units, the relevant public health departments can be contacted. National and subnational estimates of various epidemiological metrics are also available from the Global Burden of Disease (GBD) project. Additionally, the annual reports of nonprofit organizations like the Population Reference Bureau and the American Cancer Society may include up-to-date statistical estimates and projections for relevant population groups. Published scientific articles are often the best place to find very specific information about health-related exposures and diseases.

Although statistics may be readily found on the Internet, few are supported by citations and details about who collected the original data, how and when the data were collected, and how key variables were defined. When possible, trace the statistic back to its original source rather than relying on secondary reports. If the source of data is not clear, the statistic may not be trustworthy.

3.3 Abstract Databases

An **abstract** is a one-paragraph summary of an article, chapter, or book. Abstracts for journal articles in the health sciences usually provide a brief description of the study methods, the study population (such as the sample size and the location of the study), the key findings of the study, and the main conclusion or implication of the investigation. An **abstract database** is a collection of abstracts that allows researchers to search for articles using keywords or other search terms. Online databases allow researchers to quickly search thousands or even millions of abstracts for relevant articles. A careful and comprehensive

search of at least one major abstract database is the most important component of a literature search.

Some health abstract databases are available to the public at no cost (**Figure 3-1**). The most popular publicly available health science database is **PubMed**, a service of the U.S. National Library of Medicine that provides access to nearly 30 million abstracts of journal articles. Open-access databases are also available for other world regions. European PubMed Central (Europe PMC) is similar to PubMed but has more extensive coverage of European and Canadian journals. SciELO (the Scientific Electronic Library Online) and LILACS (Literatura Latino Americana e do Caribe em Ciências da Saúde) primarily focus on literature from Central and South America, and they allow searches to be conducted in English, Spanish, and Portuguese. African Journals Online (AJOL) allows searches of partner journals published by African institutions. Other national and regional abstract databases allow for searches to be conducted in other languages.

Many additional tools for finding relevant publications are available through university libraries that pay for subscriptions. Some are discipline-specific databases like CINAHL (nursing and allied health) and PsycINFO (psychology). Some are products provided by companies that produce, manage, and distribute online journal collections, such as:

- EBSCO
- JSTOR
- LexisNexis (which focuses on business and law)
- Ovid (a product of Wolters Kluwer)
- ProQuest

Some large publishing companies also offer databases of the articles published in the journals they own or publish, such as:

- Elsevier, which owns ScienceDirect
- Lippincott Williams & Wilkins (LWW), which is owned by Wolters Kluwer
- SAGE Publishing

FIGURE 3-1 Examples of Abstract Databases

Database	Description	Status
AJOL	African Journals Online	Open
CAB Direct	a product of the Centre for Agriculture and Biosciences International (CABI)	Subscription
CINAHL	Cumulative Index to Nursing and Allied Health Literature, a product of EBSCO	Subscription
Embase	a product of the large publishing company Elsevier	Subscription
ERIC	Educational Resources Information Center, a U.S. Department of Education program	Open
Europe PMC	Europe PubMed Central, similar to PubMed but with more extensive coverage of European and Canadian journals	Open
Google Scholar	a web search engine that indexes scholarly products such as book chapters, journal articles, and abstracts	Open
LILACS	Literatura Latino Americana e do Caribe em Ciências da Saúde, which focuses on literature from Central and South America	Open
MEDLINE	a U.S. National Library of Medicine database that features only journals that have applied for inclusion and passed through a review process	Open
PsycINFO	a product of the American Psychological Association (APA)	Subscription
PubMed	a service of the U.S. National Library of Medicine that includes the MEDLINE database plus additional entries (including articles published about research projects funded by the National Institutes of Health)	Open
SciELO	Scientific Electronic Library Online, which focuses on literature from Central and South America published in English, Portuguese, or Spanish	Open
Scopus	a product of the large publishing company Elsevier	Subscription
SPORTDiscus	a product from EBSCO that focuses on sports and sports medicine research	Subscription
Web of Science	a product from Clarivate Analytics that includes journals from the sciences and social sciences	Subscription

- Springer Nature
- Taylor & Francis
- Wiley-Blackwell, which owns the Wiley Online Library

A librarian can provide advice about the best databases to use for particular research questions.

Even though these health science databases cover many thousands of journals, many peer-reviewed journals are not included in any of the standard databases. Journals that are not published in English are especially likely to be omitted. A supplemental search with a

general search engine like Google Scholar may be helpful for identifying additional relevant abstracts. Additional searching is especially important when the search of discipline-specific databases yields only a small or moderate number of hits. Google Scholar has been criticized for prioritizing frequently cited articles over others that might be more relevant and for including low-quality journals that do not use rigorous peer-review processes, but it has a massive database of scholarly products. No matter what databases are searched, researchers need to assess the quality of the articles they find rather than trusting that they are rigorous simply because they have been published in a journal.

Abstract databases can be searched with keywords or MeSH terms using Boolean operators like AND, OR, and NOT. Limits can be set so that results include only abstracts with particular publication years, languages, or other selected parameters. Databases can also be searched by article title, author (often using a last name and first initials format, such as “Baker JD” or “Patel AR”), and journal title. See Chapter 26 for more information about how to successfully search abstract databases.

3.4 Full-Text Articles

Abstracts provide a glimpse into the content of an article. However, the only way to truly understand a study is to read the full text of the article. There are several ways to access the full-text versions of articles identified during a database search.

Some articles are freely available online in their entirety as open-access files on journal websites, in digital archives like PubMed Central, or on the personal websites of the authors themselves. Many open-access articles can be found by searching PubMed or other databases and looking for full-text links. For example, an article that has been added to the PMC repository will be marked as a “Free PMC Article” in PubMed. When an abstract database does not

provide a link to the full text, an Internet search tool like Google Scholar can be used to see if the full text is available in another repository.

Most university libraries subscribe to thousands of online journals that allow patrons to access electronic versions of articles. This is true even for universities in low- and middle-income countries, where research institutions can gain free access to hundreds of journals through WHO’s Hinari Access to Research in Health Programme. (Most university libraries also have a limited number of journals available in print form on their shelves, but a physical search of the stacks is unlikely to be required unless the article is relatively old.) When a particular article is not part of a university’s collection—either because the library does not subscribe to the journal or because the subscription does not cover the years or issues in which the article was published—it may be possible for a librarian to request a copy of the article from a partner institution. These “loans” of journal articles usually take the form of electronic files or photocopies of the article that do not need to be returned. Universities often offer free or low-cost inter-library loan services to affiliates, but they may restrict the number of requests that one patron can make during any given month or year.

When an article is not freely available online or through a library, another option is to contact one of the authors of the article directly and ask for a copy. The email addresses for coauthors are sometimes included in abstract databases. Sometimes they are available on the publisher’s website. For example, some subscription journals show the first page of an article as an image on their websites, and the email address for the corresponding author is often at the bottom of that page. At a minimum, many database entries and publication websites list the institutional affiliations of authors, and it is often easy to find email addresses for the authors by searching the websites of those institutions. There is no risk in writing to an author to politely request an electronic copy of an article. Most authors will be flattered that someone is interested in

their work. At worst, the requester will get no response from the author. At best, the author will send an electronic copy of the article and an offer of further assistance. Requests may also be sent through professional networking platforms, but only if the authors have active accounts with those social media sites.

3.5 Critical Reading

Once the researcher acquires a copy of the full-text article, a practical plan of action is to:

- Reread the abstract.
- Look carefully at the tables and figures, which usually display the most important results.
- Read (or at least skim) the entire text of the article.
- Review the reference list for any additional sources that should be read.

All articles should be evaluated carefully. Critical reading involves asking a series of questions about the **internal validity** of a

study, the evidence that a study measured what it intended to measure. Checks of internal validity seek to ascertain how well a particular study was designed, conducted, interpreted, and reported so that conclusions can be made about how likely it is that the resulting paper presents the truth about a particular research question in a particular population at a particular place and time (**Figure 3-2**).

Critical reading also requires questioning the **external validity** of the study, which is the likelihood that the results of a study with internal validity can be generalized to other populations, places, and times. **Generalizability** means that the results of one study are considered to be applicable to a broader target audience. Examples of questions about external validity include:

- How well do the findings of this study fit with existing knowledge about the topic? Have replication studies in diverse populations supported the generalizability of the findings?

FIGURE 3-2 Sample Questions for Critical Reading	
Section	Key Questions
Introduction/Background	What was the goal of the study? (For example, was the goal to describe a disease, examine the association between an exposure and an outcome, or test a hypothesis using an experimental design?)
Methods	Were the methods appropriate for the goal? Were the methods used to collect and analyze data scientifically valid? For example, did a study collecting new survey data use an appropriate sample population, recruit an adequate number of participants, define exposure and outcome variables appropriately, use a validated questionnaire, and apply appropriate statistical tests? Was the study conducted ethically?
Results	Was the main study question answered? Do the results seem reasonable?
Discussion	Have the authors acknowledged and discussed the limitations of the study methods? What types of bias in the design, conduct, analysis, and interpretation of the study might have caused some of the results to be inaccurate?
Conclusion	Are all of the implications and conclusions mentioned in the research report supported by the study’s results? (For example, if a study claims that an intervention caused an outcome, does the article provide sufficient evidence to support that claim of causality?)

- For experimental studies, how likely is it that the observations from the trial would occur in everyday life outside laboratory conditions?
- To what other populations might the results apply? For example, are results from a study in Canadian men ages 30 to 49 years likely to be applicable to Mexican men ages 30 to 49, Canadian women ages 30 to 49, or Canadian men ages 50 to 69?

3.6 Annotated Bibliographies

One of the common approaches for tracking the articles identified during a literature review is the creation of an **annotated bibliography**, which is a list of related publications that includes, at minimum, a full reference for each document being reviewed, a brief summary of the article or report, and a note about the resource's potential relevance to the new study. Researchers may also find it helpful to make detailed notes about how the studies featured in the annotated bibliography might inform the aims and design of the new research project. The goal is not to replicate a document's abstract. The goal is to summarize the content most pertinent to the new investigation.

Some annotated bibliographies for a new research project are compiled in document files where summary paragraphs can be typed in for each source. Sometimes it is easiest to extract the most relevant details from each source into a spreadsheet, with separate columns for various elements of the research aims, study design, study population, definitions used for exposures and outcomes, statistical results, study limitations, interpretations, and/or the reader's evaluations of internal and external validity. Alternatively, a reference management software program can allow users to add personal notes or other annotations to designated fields in a record for each source.

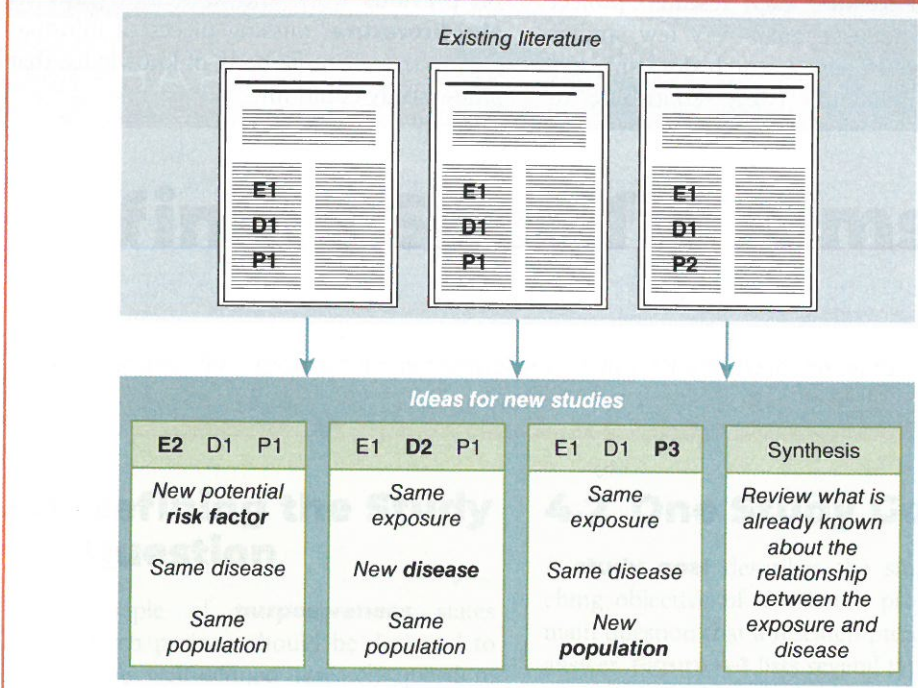
3.7 What Makes Research Original?

Every researcher is looking for an original topic to study. Originality can be a paralyzing prospect for anyone who thinks that originality requires the discovery of a newly emergent disease in a previously unrecognized group of people who live on a remote island. Such remarkable discoveries are occasionally featured in the news, but even a cursory review of the literature proves that the vast majority of original research is far less dramatic. **Originality** describes the aspects of a new research project that are novel and will allow it to make a unique contribution to the health science literature. For a research project to demonstrate originality, it needs to have one substantive difference from previous work. That could be a new exposure of interest, a new disease of interest, a new source population, a new time period under study, or a new perspective on a field of exploration. It does not require a new exposure and a new disease and a new population.

Figure 3-3 illustrates the concept of originality. An original research project could look at a new potential risk factor (E_2) for a disease (D_1) that is already well studied in a population (P_1). It could look at whether an exposure (E_1) that is known to increase the risk of one disease (D_1) in a population (P_1) also increases the risk of a second disease (D_2). It could see whether the association between an exposure (E_1) and a disease (D_1) observed in one or more parts of the world (P_1 and P_2) is also true in another part of the world (P_3). A research project using a meta-analysis approach could aim to synthesize everything that has already been published on the association between an exposure (E_1) and an outcome (D_1).

For example, a literature review might find several studies showing that older adults (the population) who take 30-minute walks several times a week (the exposure) score higher on memory tests (the outcome) than

FIGURE 3-3 Ideas for New Studies



adults who do not routinely walk for exercise. A proposed new study could ask:

- Is playing table tennis (a new exposure) effective at improving memory in older adults (the same outcome and population)?
- Do older adults who walk several times a week (the same exposure and population) improve their balance (a new outcome)?
- Does walking (the same exposure) improve memory (the same outcome) in children (a new population)?

Once a researcher identifies a possibly novel research question, a more complete review of the literature can help confirm that the area has not already been examined.

It is even possible for a replication study to be considered original research. **Replicability** means that a study protocol implemented in a

new study population should generate results similar to those of the original study, as long as the exact same protocol is used, including the same recruiting methods and inclusion criteria. **Replication studies** repeat a study protocol in a new population as part of attempting to confirm that the original findings were not due to chance. Replication studies use the same protocol for collecting data about selected exposures and outcomes that a previous study used, but they implement the study in a new population. This is an important way to determine whether observations in one population are generalizable to other populations.

Some new investigators struggle to identify a research topic that has not been previously explored in the literature, but a recognition that most research is about incremental steps forward opens up infinite options for new explorations. The main

challenge when selecting a research question is the need to limit each research project to just one focused area. Very few studies explore entirely new research domains, but every research project can contribute to

advancing a field of research when it builds on previous work and addresses **gaps in the literature**, missing pieces of information in the scientific body of knowledge that a new study could fill.